

Class 3: Simple Harmonic Motion (1)

Summer Break Bridge Course

Periodic Motion

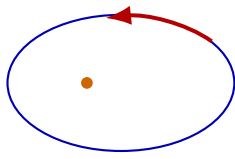
A motion is called **periodic** if after a fixed interval of time, the motion repeats itself. If $x(t)$ denotes the displacement at time t , then periodicity means

$$x(t + T) = x(t),$$

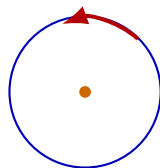
where T is called the **period**. This is the basic mathematical feature of periodic motion.

Examples of periodic motion include:

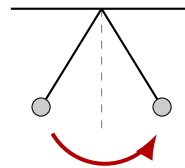
- orbital motion such as circular or elliptical motion,
- oscillation or vibration about an equilibrium position,
- linear to-and-fro motion such as a spring-block system.



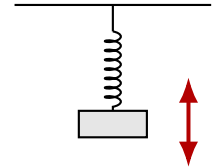
Elliptical
Orbital Periodic Motion



Circular
Orbital Periodic Motion



Oscillation / vibration
about equilibrium

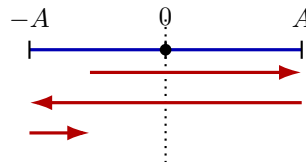
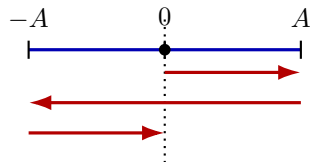


Spring-block
Linear Periodic Motion

Oscillation

An **oscillation** is a to-and-fro motion about an equilibrium position.

A **complete oscillation** means that the particle returns to the same state of motion again — that is, to the same position *and* moving in the same direction. In a crude sense, it is one full back-and-forth cycle.



Important quantities:

- **Period** T : time required to complete one full oscillation,
- **Frequency** ν : number of complete oscillations per unit time,

$$\nu = \frac{1}{T},$$

- **Amplitude A :** the maximum displacement from equilibrium on either side.

Every simple harmonic motion is oscillatory and periodic, but not every periodic motion is a simple harmonic motion.

Mechanism Behind Oscillatory Motion

Why do oscillations happen?

1. An oscillating system has an **equilibrium position**.
2. At equilibrium, the net force on the particle is zero.
3. If the particle is also at rest there, then it will remain there, as its velocity and acceleration (=force/mass) are both zero.
4. Now suppose some external agent displaces the system slightly from equilibrium.
5. A **restoring force** then starts acting, trying to pull the system back toward equilibrium.
6. This restoring force creates acceleration, so the particle moves toward equilibrium.
7. But when the particle reaches equilibrium, its velocity is generally not zero.
8. Because of this nonzero velocity, it overshoots the equilibrium position.
9. Then the restoring force again acts in the opposite direction.
10. This process repeats, producing oscillation.

At equilibrium the force becomes zero, but the motion need not stop there. The particle crosses equilibrium because of inertia.

Simple Harmonic Motion

A special and very important kind of oscillatory motion is **Simple Harmonic Motion** (SHM).

In SHM, the restoring force is:

- proportional to the displacement from equilibrium,
- and opposite in direction to the displacement.

So, if x denotes the displacement from equilibrium, then

$$F \propto -x.$$

Introducing the proportionality constant $k > 0$, we get

$$F = -kx.$$

This is the familiar **Hooke's law**. Here k is called the **force constant** or **spring constant**.

The minus sign is physically very important. It tells us that the force always points toward the equilibrium position.

Equation of Motion of SHM

Let m be the mass of the particle. By Newton's second law,

$$F = ma.$$

Since acceleration is the second derivative of displacement,

$$a = \frac{d^2x}{dt^2}.$$

Using the restoring force $F = -kx$, we get

$$m \frac{d^2x}{dt^2} = -kx.$$

Therefore,

$$\frac{d^2x}{dt^2} = -\frac{k}{m}x.$$

We now define

$$\omega^2 = \frac{k}{m},$$

so that the differential equation becomes

$$\boxed{\frac{d^2x}{dt^2} = -\omega^2x.}$$

This is the standard differential equation of SHM.

Nature of the Differential Equation

The equation

$$\frac{d^2x}{dt^2} = -\omega^2x$$

is a:

- second-order,
- linear,
- homogeneous

Ordinary Differential Equation(ODE).

Because it is second-order, its most general solution must contain **two independent constants**. Equivalently, to determine the exact motion, we need **two independent initial conditions**.

Usually, these are chosen as:

$$x(0) = x_0, \quad v(0) = v_0.$$

A single function $x(t)$ is not enough to accommodate an arbitrary initial position with an arbitrary initial velocity. Otherwise, once the position is fixed, the velocity would automatically become fixed, in a way that is too restrictive.

General Solution

Suppose $f_1(t)$ and $f_2(t)$ are two solutions of

$$\frac{d^2x}{dt^2} = -\omega^2x.$$

Then, both functions will satisfy the differential equation on their own. So,

$$\frac{d^2f_1}{dt^2} = -\omega^2f_1, \quad \frac{d^2f_2}{dt^2} = -\omega^2f_2.$$

Adding them,

$$\frac{d^2}{dt^2}(f_1 + f_2) = -\omega^2(f_1 + f_2).$$

So $f_1 + f_2$ is also a solution.

More generally, for arbitrary constants c_1, c_2 ,

$$x(t) = c_1f_1(t) + c_2f_2(t)$$

is also a solution. This is a basic property of linear homogeneous differential equations.

Hence, once two linearly independent solutions are known, the most general solution can be built from them.

Form of the General Solution for SHM

Now we ask: what kinds of functions have the property that their second derivative is proportional to minus themselves?

Sine and cosine do:

$$\begin{aligned}\frac{d^2}{dt^2} \sin(\omega t) &= -\omega^2 \sin(\omega t), \\ \frac{d^2}{dt^2} \cos(\omega t) &= -\omega^2 \cos(\omega t).\end{aligned}$$

So two independent solutions are

$$\sin(\omega t), \quad \cos(\omega t).$$

Therefore the general solution of SHM is

$$\boxed{x(t) = A_1 \sin(\omega t) + B_1 \cos(\omega t),}$$

where A_1 and B_1 are constants determined by initial conditions.

Physical significance of A_1, B_1

Differentiate:

$$x(t) = A_1 \sin(\omega t) + B_1 \cos(\omega t).$$

Then

$$v(t) = \frac{dx}{dt} = A_1\omega \cos(\omega t) - B_1\omega \sin(\omega t).$$

At $t = 0$,

$$\begin{aligned}x(0) &= A_1 \sin 0 + B_1 \cos 0 = B_1, \\v(0) &= A_1 \omega \cos 0 - B_1 \omega \sin 0 = A_1 \omega.\end{aligned}$$

So:

$$\boxed{B_1 = x(0), \quad A_1 = \frac{v(0)}{\omega}.}$$

So, in the general solution, A_1 represents the initial velocity divided by ω , and B_1 represents the initial position. Hence the solution can also be written directly in terms of initial position and initial velocity:

$$\boxed{x(t) = x(0) \cos(\omega t) + \frac{v(0)}{\omega} \sin(\omega t).}$$

Another Compact Form of the Solution

The general solution

$$x(t) = A_1 \sin(\omega t) + B_1 \cos(\omega t)$$

This can be further simplified as

$$\begin{aligned}x(t) &= A_1 \sin(\omega t) + B_1 \cos(\omega t) \\&= \sqrt{A_1^2 + B_1^2} \left[\frac{A_1}{\sqrt{A_1^2 + B_1^2}} \sin(\omega t) + \frac{B_1}{\sqrt{A_1^2 + B_1^2}} \cos(\omega t) \right]\end{aligned}$$

We now define

$$A = \sqrt{A_1^2 + B_1^2}, \quad \cos \alpha = \frac{A_1}{\sqrt{A_1^2 + B_1^2}}$$

Then,

$$\begin{aligned}\sin \alpha &= \sqrt{1 - \cos^2 \alpha} \\&= \sqrt{1 - \frac{A_1^2}{A_1^2 + B_1^2}} \\&= \frac{B_1}{\sqrt{A_1^2 + B_1^2}}\end{aligned}$$

So, finally,

$$\begin{aligned}x(t) &= \sqrt{A_1^2 + B_1^2} \left[\frac{A_1}{\sqrt{A_1^2 + B_1^2}} \sin(\omega t) + \frac{B_1}{\sqrt{A_1^2 + B_1^2}} \cos(\omega t) \right] \\&= A [\cos \alpha \sin(\omega t) + \sin \alpha \cos(\omega t)] \\&= A \sin(\omega t + \alpha)\end{aligned}$$

So, the solution can be written in a more compact form as

$$\boxed{x(t) = A \sin(\omega t + \alpha).}$$

where A and α are constants, with

$$A = \sqrt{A_1^2 + B_1^2},$$

and,

$$A_1 = A \cos \alpha, \quad B_1 = A \sin \alpha$$
$$\tan \alpha = \frac{B_1}{A_1}$$

Even in the compact form, there are still two independent constants: A and α . Nothing has been lost.

Physical Meaning of A and α

Consider

$$x(t) = A \sin(\omega t + \alpha).$$

Let

$$\theta = \omega t + \alpha.$$

Then

$$x(t) = A \sin \theta.$$

Since

$$-1 \leq \sin \theta \leq 1,$$

we have

$$-A \leq x(t) \leq A.$$

Therefore:

- A is the **amplitude** of oscillation,
- α is the **phase constant**, which tells us where in the oscillation cycle the motion starts.

At $t = 0$,

$$x(0) = A \sin \alpha.$$

So α encodes information about the initial position relative to the amplitude.

Examples: Determining the Constants

Example 1

A particle is displaced by 5 units from equilibrium and then released from rest.

So,

$$x(0) = 5, \quad v(0) = 0.$$

Using

$$x(t) = A_1 \sin(\omega t) + B_1 \cos(\omega t),$$

we know

$$B_1 = x(0) = 5, \quad A_1 = \frac{v(0)}{\omega} = 0.$$

Hence

$$x(t) = 5 \cos(\omega t).$$

Example 2

A particle is displaced by 5 units, but it is given some initial velocity so that the amplitude becomes 10 units.

Now use

$$x(t) = A \sin(\omega t + \alpha).$$

Here, the amplitude is already known:

$$A = 10.$$

Also,

$$x(0) = 5.$$

So

$$10 \sin \alpha = 5,$$

which gives

$$\sin \alpha = \frac{1}{2}.$$

Thus, one possible choice is

$$\alpha = \frac{\pi}{6}.$$

Hence,

$$x(t) = 10 \sin \left(\omega t + \frac{\pi}{6} \right).$$

If the particle is released from rest at a displacement $x(0)$, then the amplitude equals $|x(0)|$. But if the particle is given a nonzero initial velocity, then the amplitude can be larger than the initial displacement.

Velocity as a Function of Time

From

$$x(t) = A \sin(\omega t + \alpha),$$

we get

$$v(t) = \frac{dx}{dt} = A\omega \cos(\omega t + \alpha).$$

So the maximum possible speed is

$$|v|_{\max} = A\omega.$$

Relation Between Velocity and Position

Start from

$$x(t) = A \sin(\omega t + \alpha), \quad v(t) = A\omega \cos(\omega t + \alpha).$$

Let

$$\theta = \omega t + \alpha.$$

Then

$$x = A \sin \theta, \quad v = A\omega \cos \theta.$$

Using

$$\sin^2 \theta + \cos^2 \theta = 1,$$

we get

$$\frac{x^2}{A^2} + \frac{v^2}{A^2 \omega^2} = 1.$$

Hence

$$\boxed{v = \pm \omega \sqrt{A^2 - x^2}.}$$

This equation shows:

- at $x = \pm A$, the speed is zero,
- at $x = 0$, the speed is maximum,
- for a fixed amplitude A , position and velocity lie on an ellipse in the (x, v) plane.

Acceleration and Its Relation with Position

Differentiate velocity:

$$a(t) = \frac{dv}{dt} = \frac{d^2 x}{dt^2}.$$

Since

$$x(t) = A \sin(\omega t + \alpha),$$

we have

$$a(t) = -A\omega^2 \sin(\omega t + \alpha).$$

But

$$x(t) = A \sin(\omega t + \alpha),$$

so

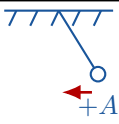
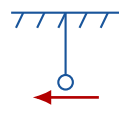
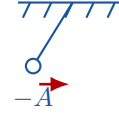
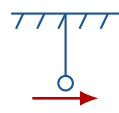
$$\boxed{a(t) = -\omega^2 x(t).}$$

This is the defining dynamical feature of SHM: acceleration is always opposite to displacement and proportional to it.

The maximum magnitude of acceleration is

$$|a|_{\max} = \omega^2 A.$$

Important Extremes in SHM

Schematic Location	Position(x)	Velocity(v)	Acceleration(a)
	$+A$	0	$-\omega^2 A$
	0	$-A\omega$	0
	$-A$	0	$+\omega^2 A$
	0	$+A\omega$	0

At the extremes, speed is zero but the restoring acceleration has maximum magnitude. At equilibrium, acceleration is zero but the speed has maximum magnitude.

Final Perspective

Simple harmonic motion is one of the most fundamental motions in physics. Its importance comes from three connected ideas:

- the restoring force is proportional to displacement and opposite in direction,
- the equation of motion is a second-order linear homogeneous differential equation,
- the resulting motion is sinusoidal in time.

So SHM connects:

physical mechanism $\rightarrow$ equation of motion $\rightarrow$ sinusoidal solution $\rightarrow$ clear relations among x , v , a , T , ω

Key Takeaway:

In simple harmonic motion, the farther the particle is from equilibrium, the more strongly it is pulled back. This leads to a motion that is periodic, sinusoidal, and completely determined by two independent constants.